TUGAS 03 REVISI

CheatSheet Bab 3

Matakuliah Workshop DevOps

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September 2026

1. Membuat directory praktikum

```
toimul@DESKTOP-2OIAP1F:~/docker-lab$ mkdir -p ~/docker-lab/bab-3/app
toimul@DESKTOP-2OIAP1F:~/docker-lab$ mkdir -p ~/docker-lab/bab-3/html
toimul@DESKTOP-2OIAP1F:~/docker-lab$ cd ~/docker-lab/bab-3
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$
```

2. Menulis compose.yaml

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ code compose.yaml
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$
```

compose.yaml

Ubuntu > home > toimul > docker-lab > bab-3 > compose.yaml

```yaml
services:
  web:
    image: nginx:alpine
    ports:
      - "127.0.0.1:8080:80"
    volumes:
      - ./html:/usr/share/nginx/html:ro
      - ./nginx.conf:/etc/nginx/conf.d/default.conf:ro
    networks:
      - frontend
    depends_on:
      - app

  app:
    build:
      context: ./app
    environment:
      DB_HOST: db
      DB_PORT: "5432"
      DB_NAME: labdb
      DB_USER: labuser
      DB_PASS: labpass123
    networks:
      - frontend
      - backend
    depends_on:
      db:
        condition: service_healthy

  db:
    image: postgres:16-alpine
    environment:
      POSTGRES_DB: labdb
      POSTGRES_USER: labuser
      POSTGRES_PASSWORD: labpass123
    volumes:
      - pg-data:/var/lib/postgresql/data
    networks:
      - backend
    healthcheck:
      test:
        - CMD-SHELL
        - pg_isready -U labuser -d labdb
      interval: 5s
      timeout: 5s
      retries: 5

volumes:
  pg-data:

networks:
  frontend:
  backend:
```

3. Membuat aplikasi flask

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ code app/requirements.txt
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$
```

Ubuntu > home > toimul > docker-lab > bab-3 > app > requirements.txt

```
Flask==3.1.2
psycopg[binary]==3.2.9
gunicorn==23.0.0
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ code app/app.py
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$
```

Ubuntu > home > toimul > docker-lab > bab-3 > app > app.py > health

```python
import os

import psycopg
from flask import Flask, jsonify

app = Flask(__name__)


def database_connection():
    return psycopg.connect(
        host=os.environ["DB_HOST"],
        port=os.getenv("DB_PORT", "5432"),
        dbname=os.environ["DB_NAME"],
        user=os.environ["DB_USER"],
        password=os.environ["DB_PASS"],
    )


@app.get("/")
def index():
    try:
        with database_connection() as connection:
            with connection.cursor() as cursor:
                cursor.execute("SELECT version();")
                database_version = cursor.fetchone()[0]

        return jsonify(
            status="ok",
            message="Nginx, Flask, dan PostgreSQL berhasil terhubung",
            database=database_version,
        )
    except Exception as error:
        return jsonify(
            status="error",
            message="Koneksi database gagal",
            detail=str(error),
        ), 500


@app.get("/health")
def health():
    try:
        with database_connection() as connection:
            with connection.cursor() as cursor:
                cursor.execute("SELECT 1;")
                cursor.fetchone()

        return jsonify(status="healthy"), 200
    except Exception:
        return jsonify(status="unhealthy"), 503
```

4. Membuat Dockerfile aplikasi

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ code app/Dockerfile
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$
```

Ubuntu > home > toimul > docker-lab > bab-3 > app > Dockerfile

```dockerfile
FROM python:3.12-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

RUN useradd --system --uid 10001 appuser
USER appuser

EXPOSE 5000

CMD ["gunicorn", "--bind", "0.0.0.0:5000", "--workers", "2", "app:app"]
```

5. Membuat konfigurasi Nginx

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ code nginx.conf
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$
```

Ubuntu > home > toimul > docker-lab > bab-3 > nginx.conf

```
upstream flask_backend {
    server app:5000;
}

server {
    listen 80;
    server_name localhost;

    location / {
        proxy_pass http://flask_backend;
        proxy_http_version 1.1;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        proxy_connect_timeout 5s;
        proxy_read_timeout 30s;
    }

    location = /static.html {
        root /usr/share/nginx/html;
    }
}
```

6. Membuat halaman statis optional

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ code html/index.html
```

Ubuntu > home > toimul > docker-lab > bab-3 > html > index.html > html > body > p

```
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <title>Laboratorium Docker Compose</title>
</head>
<body>
    <h1>Docker Compose Bab 3</h1>
    <p>Bind mount Nginx berhasil digunakan.</p>
</body>
</html>
```

7.1. Memvalidasi Compose

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ docker compose config
name: bab-3
services:
  app:
    build:
      context: /home/toimul/docker-lab/bab-3/app
      dockerfile: Dockerfile
    depends_on:
      db:
        condition: service_healthy
        required: true
    environment:
      DB_HOST: db
      DB_NAME: labdb
      DB_PASS: labpass123
      DB_PORT: "5432"
      DB_USER: labuser
    networks:
      backend: null
      frontend: null
  db:
    environment:
      POSTGRES_DB: labdb
      POSTGRES_PASSWORD: labpass123
      POSTGRES_USER: labuser
    healthcheck:
      test:
        - CMD-SHELL
        - pg_isready -U labuser -d labdb
```

7.2. Memeriksa service yang akan dibuat

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ docker compose config --services
db
app
web
```

8.1. Menjalankan stack

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ docker compose up -d --build
[+] Building 22.4s (9/12)
 => [internal] load local bake definitions                                    0.0s
 => => reading from stdin 502B                                                0.0s
 => [internal] load build definition from Dockerfile                          0.0s
 => => transferring dockerfile: 320B                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim           3.6s
 => [auth] library/python:pull token for registry-1.docker.io                 0.0s
 => [internal] load .dockerignore                                             0.0s
 => => transferring context: 2B                                               0.0s
 => [1/6] FROM docker.io/library/python:3.12-slim@sha256:2f17fc044b579bab302c2e8054d3a686e2cb9a83de48e70534b94cd8  0.1s
 => => resolve docker.io/library/python:3.12-slim@sha256:2f17fc044b579bab302c2e8054d3a686e2cb9a83de48e70534b94cd8  0.1s
 => [internal] load build context                                             0.0s
 => => transferring context: 1.45kB                                           0.0s
 => CACHED [2/6] WORKDIR /app                                                 0.0s
 => [3/6] COPY requirements.txt .                                             0.1s
 => [4/6] RUN pip install --no-cache-dir -r requirements.txt                 18.2s
 => => # (22 kB)
 => resolving provenance for metadata file                                    0.0s
[+] up 7/7
 ✔Image bab-3-app          Built                                             33.9s
 ✔Network bab-3_backend    Created                                            0.1s
 ✔Network bab-3_frontend   Created                                            0.1s
 ✔Volume bab-3_pg-data     Created                                            0.0s
 ✔Container bab-3-db-1     Healthy                                           11.9s
 ✔Container bab-3-app-1    Started                                           12.5s
 ✔Container bab-3-web-1    Started                                           12.6s
```

8.2. Periksa status

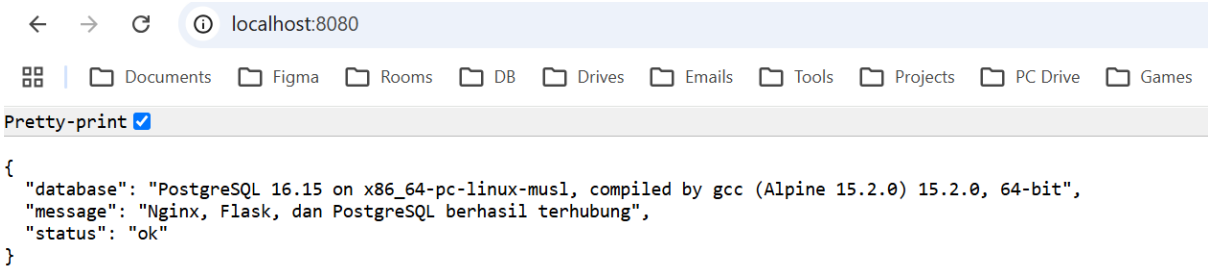
```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ docker compose ps
NAME          IMAGE               COMMAND                  SERVICE   CREATED         STATUS                       PORT
S
bab-3-app-1   bab-3-app           "gunicorn --bind 0.0…"   app       2 minutes ago   Up About a minute            5000
/tcp
bab-3-db-1    postgres:16-alpine  "docker-entrypoint.s…"   db        2 minutes ago   Up About a minute (healthy)  5432
/tcp
bab-3-web-1   nginx:alpine        "/docker-entrypoint.…"   web       2 minutes ago   Up About a minute            127.
0.0.1:8080->80/tcp
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$
```

9. Menguji aplikasi

9.1. Via curl

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ curl http://localhost:8080/
{"database":"PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit","message":"Nginx,
 Flask, dan PostgreSQL berhasil terhubung","status":"ok"}
```

9.2. Via browser



9.3. Uji health endpoint

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ curl -i http://localhost:8080/health
HTTP/1.1 200 OK
Server: nginx/1.31.5
Date: Tue, 22 Sep 2026 14:53:11 GMT
Content-Type: application/json
Content-Length: 21
Connection: keep-alive

{"status":"healthy"}
```

9.4. Uji halaman statis

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ curl http://localhost:8080/static.html
<html>
<head><title>404 Not Found</title></head>
<body>
<center><h1>404 Not Found</h1></center>
<hr><center>nginx/1.31.5</center>
</body>
</html>
```

10.1. Memeriksa log

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-3$ docker compose logs --tail 100
app-1  | [2026-09-22 14:48:06 +0000] [1] [INFO] Starting gunicorn 23.0.0
db-1   | The files belonging to this database system will be owned by user "postgres".
db-1   | This user must also own the server process.
db-1   |
db-1   | The database cluster will be initialized with locale "en_US.utf8".
db-1   | The default database encoding has accordingly been set to "UTF8".
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
```

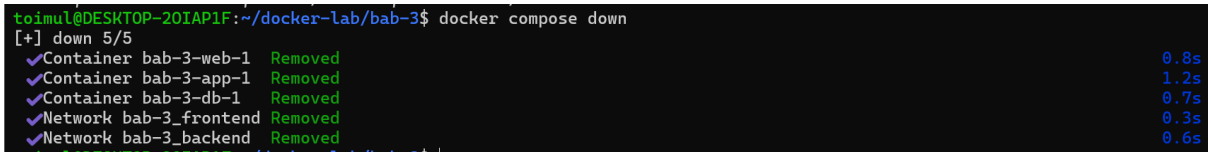
10.2. docker compose logs -f => memeriksa log secara real-time

10.3. Memeriksa log tertentu

- docker compose logs app
- docker compose logs db
- docker compose logs web

11. Menghentikan lingkungan

11.1. Menghapus data container dan network, tetapi masih mempertahankan data postgreSql



11.2. Menghapus container, network, sekaligus volume PostgreSQL

```
docker compose down -v
```